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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	FOOD DELIVERY TRACKING APPLICATION	
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Abstract:

Online food ordering has become an integral part of daily life across the world, with customers expecting speed, transparency, and reliability from the moment an order is placed until it arrives at their doorstep. A recurring pain point in this ecosystem is the lack of accurate, real-time visibility into an order's journey: customers are often left guessing about preparation delays, pickup times, and the delivery agent's exact location, while restaurants and delivery platforms struggle to coordinate orders, riders, and routes efficiently. Existing tracking mechanisms in many small and mid-sized platforms rely on static status updates (such as "Order Placed", "Out for Delivery", "Delivered") rather than continuous, live location tracking, which reduces customer trust and increases support overhead. This highlights the need for a dedicated Food Delivery Tracking Application that unifies order management, live GPS-based tracking, and timely notifications into a single, reliable system for customers, delivery agents, and restaurant/admin staff.

To address this need, the proposed Food Delivery Tracking Application is designed as a multi-role platform built using a client-server architecture, with a mobile/web front end for customers and delivery agents, a RESTful backend for business logic, and a relational/NoSQL database for persisting

order and location data. The system follows a service-oriented approach, structured around distinct modules for user management, restaurant and menu management, order processing, live tracking, and notifications, so that each concern can be developed, scaled, and maintained independently.

The first module covers user registration and authentication for customers, delivery agents, and restaurant administrators, along with restaurant and menu management, allowing restaurants to list items, update availability, and manage pricing in real time.

The order-processing module handles cart management, order placement, and payment confirmation, automatically assigning an available delivery agent based on proximity and current workload. Once an order is confirmed, the system moves it through a defined lifecycle of states, from order placed and accepted, to preparation, pickup, and delivery, with every transition logged and time-stamped for accountability.

The core of the system is the real-time tracking module, which continuously captures the delivery agent's GPS coordinates from their mobile device and streams these updates to the customer's app using WebSocket or push-based communication so the position on the map refreshes live rather than on manual refresh. The module also computes the estimated time of arrival (ETA) using the agent's current location, route distance, and average travel speed, updating the estimate dynamically as conditions change.

A complementary notification module keeps every stakeholder informed through push notifications and status alerts at each stage of the order lifecycle, while an admin dashboard gives restaurant and platform staff a consolidated view of active orders, delivery agent availability, and delivery performance metrics such as average delivery time and on-time delivery rate.

Collectively, these modules come together to form an end-to-end Food Delivery Tracking Application that improves transparency for customers, streamlines coordination for restaurants and delivery agents, and gives administrators actionable operational insight. By combining live location tracking, dynamic ETA calculation, and proactive notifications, the proposed system aims to reduce customer uncertainty, minimise delivery-related support queries, and improve overall service reliability in the food delivery ecosystem.

Project Architecture/Project Overview:

